

Algebra & Polynomials

Like Terms · Substitution · Polynomials · Monomial Products & Quotients · Exponent Laws

MTH1W · Class — Guided Practice

Name: _____

Date: _____

Goal: Simplify, evaluate, add, subtract, multiply and divide polynomials using exponent laws.

Warm-up (quick recall)

1. Simplify: $4a + 6a$.

2. State the coefficient of the term $-5x^2$.

3. Simplify: $7m - m$.

Examples & Practice

For each skill, study the worked **Example**, then solve the questions that follow. Difficulty rises within each set (L1 → L3). Answers are on the last page.

Collecting Like Terms

Example:

Simplify $5x + 3y - 2x + 4y$.

Group like terms: $(5x - 2x) + (3y + 4y)$.

$= 3x + 7y$. Answer: $3x + 7y$.

4. Simplify: $8a + 3a - 2a$. **[L1]**

5. Simplify: $6x + 4 - 2x + 5$. **[L1]**

6. Simplify: $3m - 5n + 7m + 2n - 4$. **[L2]**

7. A student wrote $4x + 3x^2 = 7x^3$. Explain the error and give the correct simplification. **[L3]**

Evaluating by Substitution

Example:

Evaluate $3x - 2$ when $x = 4$.

Substitute: $3(4) - 2 = 12 - 2$.

$= 10$. Answer: 10.

8. Evaluate $2x + 5$ when $x = 3$. **[L1]**

9. Evaluate $4a - b$ when $a = 2$ and $b = -3$. **[L2]**

10. Evaluate $x^2 - 3x$ when $x = 5$. **[L2]**

11. The cost of renting a hall is $C = 15n + 40$, where n is the number of hours. Find the cost for 6 hours and state what the 40 represents. **[L3]**

Adding & Subtracting Polynomials

Example:

Simplify $(3x^2 + 2x - 1) + (x^2 - 5x + 4)$.

Combine like terms: $(3x^2 + x^2) + (2x - 5x) + (-1 + 4)$.
 $= 4x^2 - 3x + 3$.

12. Simplify: $(2x + 3) + (5x - 1)$. **[L1]**

13. Simplify: $(4a^2 - 2a + 6) - (a^2 + 3a - 2)$. **[L2]**

14. Subtract $(2x^2 - x)$ from $(5x^2 + 3x)$. **[L2]**

15. A triangle has perimeter $(9x + 4)$. Two sides are $(3x + 1)$ and $(2x - 3)$. Find the third side. **[L3]**

Multiplying by a Constant or Monomial

Example:

Expand $3(2x - 5)$ and $2x(x + 4)$.

$$3(2x - 5) = 6x - 15. \quad 2x(x + 4) = 2x^2 + 8x.$$

Distribute the factor to every term.

16. Expand: $4(x + 3)$. **[L1]**

17. Expand: $-2(3a - 5)$. **[L1]**

18. Expand: $3x(2x - 4)$. **[L2]**

19. Expand and simplify: $5(2x - 3) - 2(x - 4)$. **[L3]**

Dividing by a Constant or Monomial

Example:

Simplify $(6x + 9) \div 3$ and $(8x^2 - 4x) \div 4x$.
 $(6x + 9) \div 3 = 2x + 3$. $(8x^2 - 4x) \div 4x = 2x - 1$.
Divide every term by the divisor.

20. Simplify: $(10x + 15) \div 5$. **[L1]**

21. Simplify: $(12a^2 - 8a) \div 4$. **[L2]**

22. Simplify: $(15x^2 + 10x) \div 5x$. **[L2]**

23. Simplify: $(6x^3 - 9x^2 + 3x) \div 3x$. **[L3]**

Exponent Laws Applied to Terms

Example:

Simplify $(2x^2)(3x^4)$ and $(x^5) \div (x^2)$.

Multiply: coefficients $2 \times 3 = 6$, add exponents $\rightarrow 6x^6$.

Divide: subtract exponents $\rightarrow x^3$.

24. Simplify: $a^3 \cdot a^4$. **[L1]**

25. Simplify: $(2m^3)(4m^2)$. **[L1]**

26. Simplify: $(12x^5) \div (3x^2)$. **[L2]**

27. Simplify: $(2a^2b)^3$. **[L3]**

Answers

Warm-up: W1) 10a W2) -5 W3) 6m

- 4.** (L1) $9a$
- 5.** (L1) $4x + 9$
- 6.** (L2) $10m - 3n - 4$
- 7.** (L3) $4x$ and $3x^2$ are not like terms (different powers), so they cannot be combined:
 $3x^2 + 4x$.
- 8.** (L1) $2(3)+5 = 11$
- 9.** (L2) $4(2) - (-3) = 8 + 3 = 11$
- 10.** (L2) $25 - 15 = 10$
- 11.** (L3) $C = 15(6) + 40 = \$130$; the 40 is the fixed base fee.
- 12.** (L1) $7x + 2$
- 13.** (L2) $3a^2 - 5a + 8$
- 14.** (L2) $3x^2 + 4x$
- 15.** (L3) $(9x+4) - (3x+1) - (2x-3) = 4x + 6$.
- 16.** (L1) $4x + 12$
- 17.** (L1) $-6a + 10$
- 18.** (L2) $6x^2 - 12x$
- 19.** (L3) $10x - 15 - 2x + 8 = 8x - 7$
- 20.** (L1) $2x + 3$
- 21.** (L2) $3a^2 - 2a$
- 22.** (L2) $3x + 2$
- 23.** (L3) $2x^2 - 3x + 1$
- 24.** (L1) a^7
- 25.** (L1) $8m^5$
- 26.** (L2) $4x^3$
- 27.** (L3) $8a^6b^3$